LAB EXPERIMENT RECORD

Set 22 Time Series Analysis

Department	AIDS
Course Code / Course Name	DSA0602/ Data Handling and Visualization
Experiment	Data Visualization using R
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Date	15/10/2026
Day	Thursaday

AIM

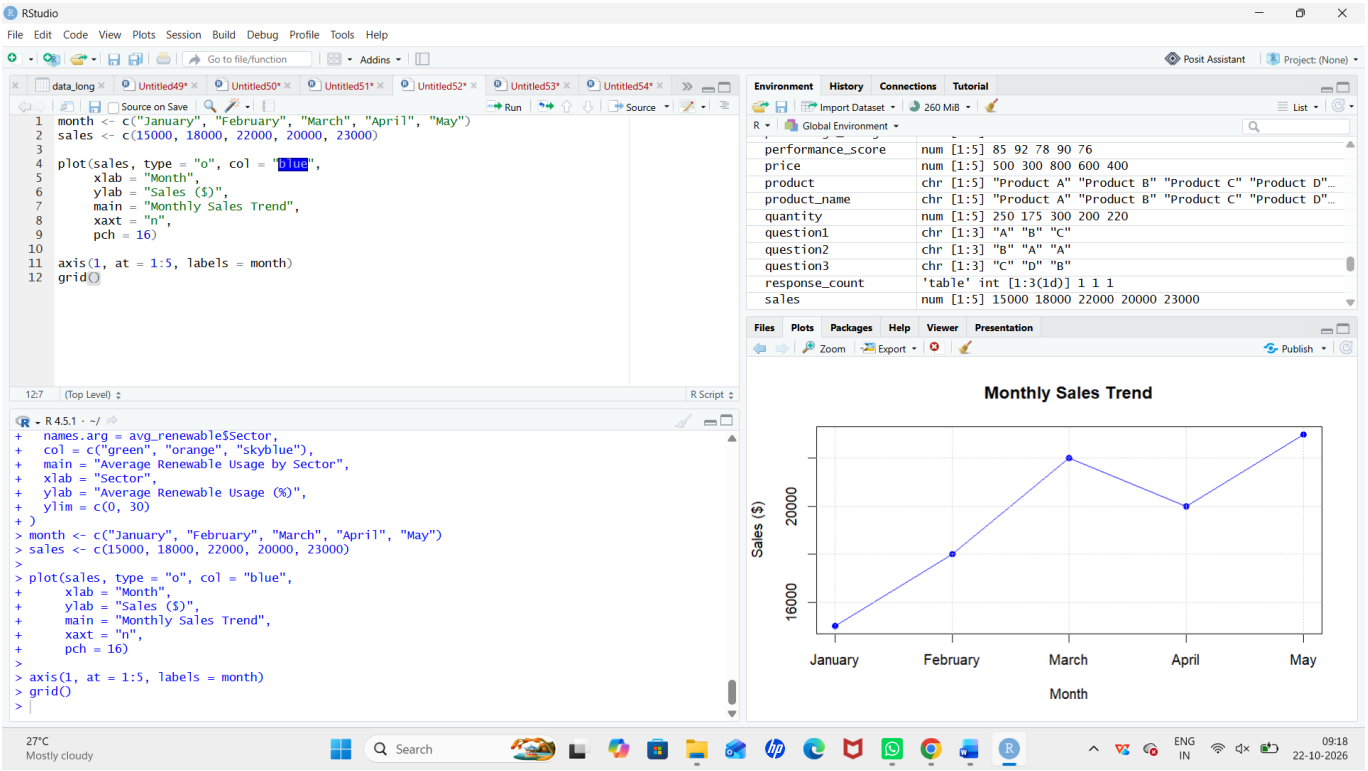
To analyze monthly sales data using R and Tableau by creating a time series line chart, a scatter plot for advertising budget versus sales, a pie chart for product-wise sales distribution, and an autocorrelation plot to examine sales patterns.

1.

```
month <- c("January", "February", "March", "April", "May")
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(sales, type = "o", col = "blue",
     xlab = "Month",
     ylab = "Sales ($)",
     main = "Monthly Sales Trend",
     xaxt = "n",
     pch = 16)

axis(1, at = 1:5, labels = month)
grid()
```

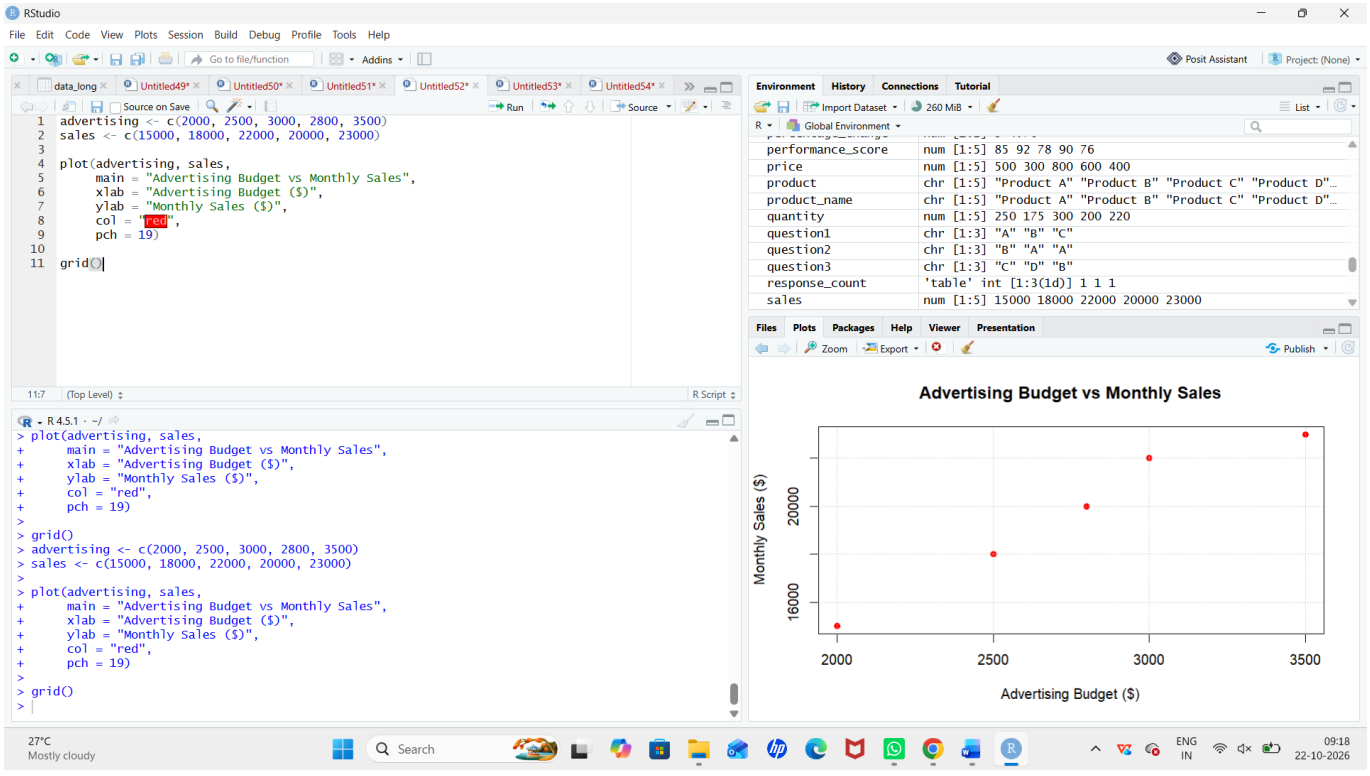


2

```
advertising <- c(2000, 2500, 3000, 2800, 3500)
sales <- c(15000, 18000, 22000, 20000, 23000)

plot(advertising, sales,
     main = "Advertising Budget vs Monthly Sales",
     xlab = "Advertising Budget ($)",
     ylab = "Monthly Sales ($)",
     col = "red",
     pch = 19)

grid()
```

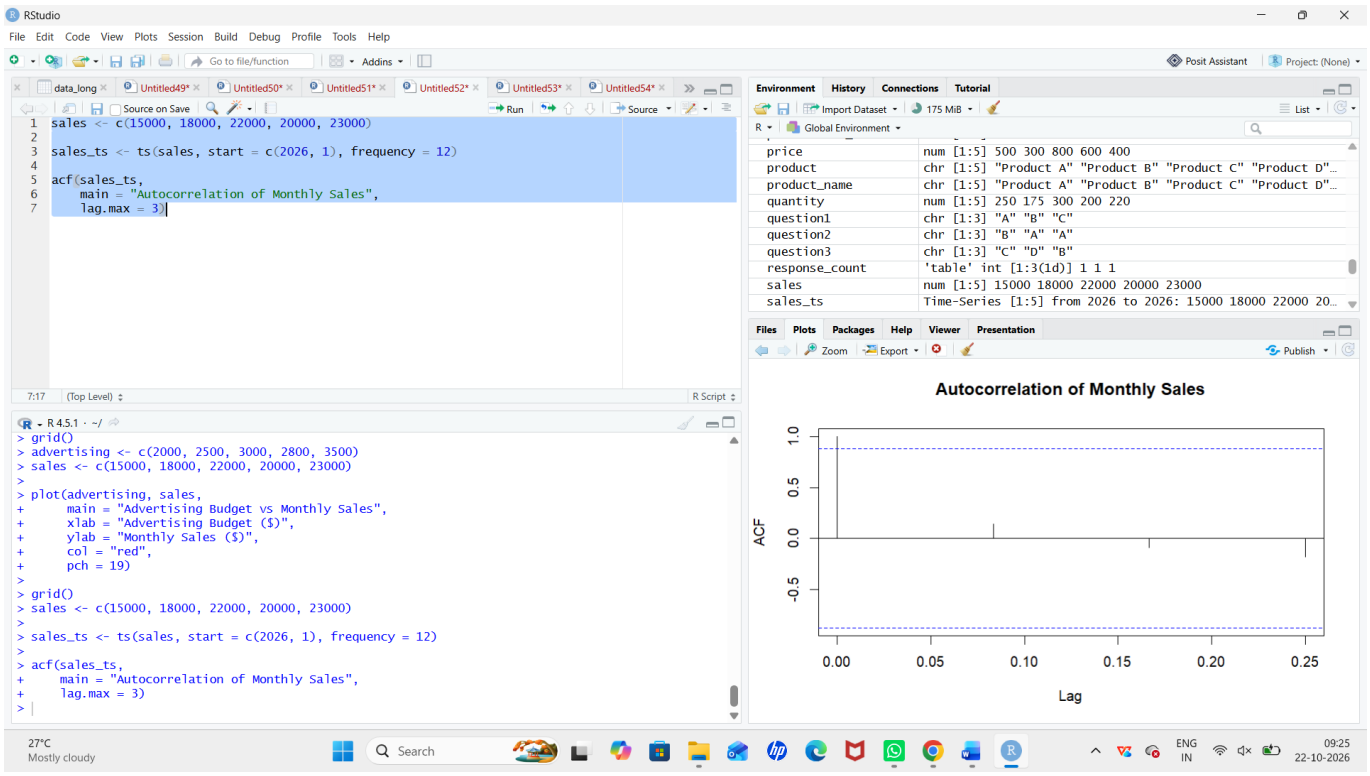


3.

```
sales <- c(15000, 18000, 22000, 20000, 23000)

sales_ts <- ts(sales, start = c(2026, 1), frequency = 12)

acf(sales_ts,
    main = "Autocorrelation of Monthly Sales",
    lag.max = 3)
```



3.

1. Prepare the monthly sales data in Excel or CSV format.
2. Open Tableau and connect to the dataset.
3. Create Sheet 1 – Line Chart by dragging Month to Columns and Sales to Rows.
4. Set the sales aggregation to SUM and select the Line chart type.
5. Create Sheet 2 – Pie Chart using a product dataset containing Product and Sales columns.
6. Drag Product to Color and Sales to Angle. Display sales values or percentages as labels.
7. Open a new Dashboard and add both sheets.
8. Adjust the layout, titles, and colors to make the dashboard clear.

Note: Product-level sales data was not provided, so you will need to add product names and their sales values to create the pie chart.

Procedure

1. Create the monthly sales dataset containing Month and Sales values from January to May.

2. Define the dataset and required variables in R.

3. Create a time series line chart to visualize monthly sales trends.

4. Generate a scatter plot to analyze the relationship between advertising budget and monthly sales using sample data.

5. Import the sales dataset into Tableau and create a line chart showing monthly sales trends.

6. Create a pie chart to display product-wise sales distribution using product-level data.

7. Combine both charts into a single Tableau dashboard.

8. Generate an autocorrelation plot in R to examine relationships between sales values at different time lags.

9. Observe and interpret the visualizations to understand sales trends and possible patterns.

Result

Monthly sales data was analyzed successfully using R and Tableau. The line chart displayed the sales trend, the scatter plot illustrated the relationship between advertising budget and sales, and the Tableau dashboard combined monthly sales trends with product-wise sales distribution. The autocorrelation plot helped examine time-series patterns. However, additional data over a longer period is required to identify seasonality reliably.